

Starvation transforms signal encoding in *C. elegans* thermoresponsive neurons and suppresses heat avoidance via bidirectional glutamatergic and peptidergic signaling

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eLife Assessment

This **important** study shows how hunger alters avoidance of harmful heat in *C. elegans* by reconfiguring the activity of key sensory neurons. The evidence is **convincing**, with well-designed behavioural, genetic, and imaging experiments that support the main conclusions. The work will be of interest to neuroscientists studying how internal states shape sensory processing and behaviour across species.

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Abstract

Animals must continuously adapt their behavioral outputs in response to changes in internal state, including nutritional state. Here, we show that starvation induces a profound and progressive suppression of thermonociceptive behavior in *Caenorhabditis elegans*. The thermoresponsive sensory neurons AWCs mediate robust heat-evoked reversals over a broad range of stimulus intensities via glutamate and FLP-6 neuropeptide signaling, each covering distinct heat intensity ranges. After six hours of food deprivation, heat-evoked reversal responses are nearly abolished, independently of any external food odor cues. Prolonged food deprivation triggers a switch in AWC heat-evoked activity patterns, transitioning from a predictable, stimulus-locked response mode to a heterogeneous and stochastic regime. This switch relies on ASI neurons, proposed to work as internal state-sensing neurons. INS-32 and NLP-18 neuropeptide signals from ASI switch from a reversal-promoting to a reversal-inhibiting effect. Glutamatergic transmission from non-AWC neurons is also engaged to suppress reversals. Our findings define a circuit logic by which nociceptive responsiveness gating by internal nutritional state is linked to dynamic modulation of sensory neuron activity patterns and orchestrated by bidirectional glutamatergic and neuropeptidergic signals. More broadly, this study illustrates how sensory systems integrate metabolic

information to prioritize behavioral outputs under changing physiological conditions, providing mechanistic insight into the plastic coupling between sensation, internal state, and action selection.

Introduction

To ensure survival, animals must detect and avoid noxious stimuli, such as harmful chemicals, toxic gases, or extreme temperatures. These avoidance behaviors, often categorized as nociceptive, are conserved across phyla and essential for minimizing damage and guiding appropriate behavioral responses. Like most animals studied so far, the nematode *Caenorhabditis elegans*, can produce robust innate escape behaviors and modulate these responses according to external and internal regulatory cues [1–4]. Many genes involved in mammalian pain perception and plasticity—such as the Transient Receptor Potential channels, CaM kinase and Calcineurin—are also implicated in worms, underscoring the evolutionary conservation of nociceptive pathways and the broad interest of the *C. elegans* model [5–9].

C. elegans is a small ectotherm, whose temperature equilibrates almost instantly with its surroundings. It can grow in a 15 to 25°C range. *C. elegans* produces robust avoidance behaviors in response to noxious temperature (which can be defined as temperature above 26°C) or to fast-raising thermal stimuli even below 26°C [10]. Upon acute thermal stimulation, forward-moving worms respond to head-targeted or whole animal heating by initiating backward locomotion (reversal event) and to tail-targeted heating by accelerating their forward locomotion [11]. Several thermo-responsive neurons have been characterized (see [12] for a review). Among these neurons, AFD neurons are the best characterized primary thermosensory neurons responding to temperature changes and mediating both thermotaxis in response to innocuous thermal cues and noxious heat avoidance response [13]. FLP neurons are tonic thermosensors whose activity level continuously reflects the current temperature, and which regulate animal speed and reorientation maneuvers [14, 15]. AWC neurons are polymodal sensory neurons responding to olfactory cues and to temperature in a context-dependent manner [16–18]. The pair of AWC sensory neurons asymmetrically differentiate into AWC^{ON} and AWC^{OFF}, which express specific markers and respond either similarly or differentially depending on the stimulus type [19]. Regarding AWC response to temperature change, both deterministic (stimulus-locked calcium elevations upon warming [18]), and stochastic responses, whose frequency can be modulated by warming [20], have been reported depending on experimental conditions. In addition, extremely powerful IR-laser based pulses—causing a 10°C elevation from 23 to 33°C within tens of milliseconds—were shown to trigger subtype-specific response: calcium elevation in AWC^{ON} and calcium decrease in AWC^{OFF} [21]. ASI sensory neurons have also been reported to respond to change in temperature in the innocuous range [22], but their implication in noxious-heat response is not known.

Neuromodulators, including neuropeptides and monoamines, play a central role in reconfiguring behavioral responses to sensory inputs based on context [15, 23, 24]. In particular, neuropeptides have been shown to drive behavioral plasticity across a range of paradigms in *C. elegans*, including feeding-dependent changes in locomotion and sensory responsiveness as a response to changes in the animal's internal physiological and metabolic states [25–27]. A reduction of nociceptive sensitivity or responsiveness during prolonged starvation could in principle help worms travel across harsher environments in order to find new food sources.

Consistent with this potential ecological advantage, food deprivation-dependent reduction has been shown for the response to high osmolarity and chemical repellents [1, 28]. Examples of plasticity in the thermonociceptive response of *C. elegans* are so far essentially limited to the down-regulation of noxious heat avoidance in response to persistent exposure to moderately

noxious heat level (28°C) or repeated heat stimulations [9, 29–31]. The impact of prolonged starvation and the contribution of neuromodulation in the context-dependent regulation of thermnociceptive behaviors is largely unknown.

Here, we investigate how feeding state modulates heat avoidance behavior in *C. elegans*. We show that prolonged starvation significantly suppresses thermnociceptive responses, a plasticity not driven by external chemosensory cues but caused by nutrient availability. We identify a critical role for the AWC sensory neurons in mediating heat-evoked escape responses, and for the ASI neuroendocrine neurons in modulating these responses under starvation. Our data show that AWC neurons employ both glutamate and the neuropeptide FLP-6 to mediate avoidance, with distinct contributions from AWC^{ON} and AWC^{OFF} subtypes. Under early food deprivation, AWC responses to heat are stereotyped and excitatory, but under prolonged starvation, these responses become variable, reflecting a shift from deterministic to stochastic encoding. This shift depends on the ASI neurons, which modulate heat-evoked response via NLP-18 and INS-32 neuropeptides. Additionally, our findings suggest that glutamatergic signaling from non-AWC sources also contribute to the suppression thermnociception following starvation. Together, our results suggest a neuromodulatory mechanism by which internal nutritional state reconfigures nociceptive behavior. This demonstrates that even hard-wired escape responses are not immune to physiological context, and highlights a broader principle: survival strategies in simple animals balance external threats against internal needs through circuit-level plasticity.

Results

Starvation downregulates thermnociceptive responses in *C. elegans*

To assess how the feeding state modulates thermnociceptive behavior in *C. elegans*, we compared responses across different durations of food deprivation (**Figure 1A**). Although fed animals showed high sensitivity to noxious heat, they also displayed an elevated baseline of spontaneous reversals, which limited their utility as a control group (**Figure 1B**). A 1-hour off-food condition reduced spontaneous reversals and led to a mild attenuation of heat-evoked responses at low stimulus intensities, while responses to stronger stimuli remained comparable to those of fed animals.

Prolonged food deprivation led to a striking progressive reduction in thermnociceptive responses, with responses after 6 hours of starvation approaching baseline spontaneous reversal rates (**Figure 1B and C**). This suggests a robust inhibition of nociceptive behavior. To determine whether this attenuation was due to the absence of nutrients or chemosensory cues, we conducted similar starvation experiments in the presence of food odor (**Figure 1D**). The reduction in thermnociceptive response persisted, indicating that the effect is driven by the internal starvation state rather than external olfactory input.

Based on this observation, we focused subsequent analyses on dissecting the circuit and molecular underpinnings of starvation-dependent plasticity, using the 1-hour and 6-hour off-food conditions as representative of early food deprivation and prolonged starvation, respectively.

Distinct roles of AWC and ASI neurons in heat-evoked response and starvation-dependent plasticity

To identify the neural substrates underlying thermnociceptive behavior and its modulation by starvation, we genetically ablated candidate thermosensory neurons (**Figure 2**). Surprisingly, ablation of AFD—the canonical thermosensory neuron—had no significant effect under either 1h or 6h food deprivation (**Figure 2 Supplement 1**). Simultaneous ablation of AFD and FLP

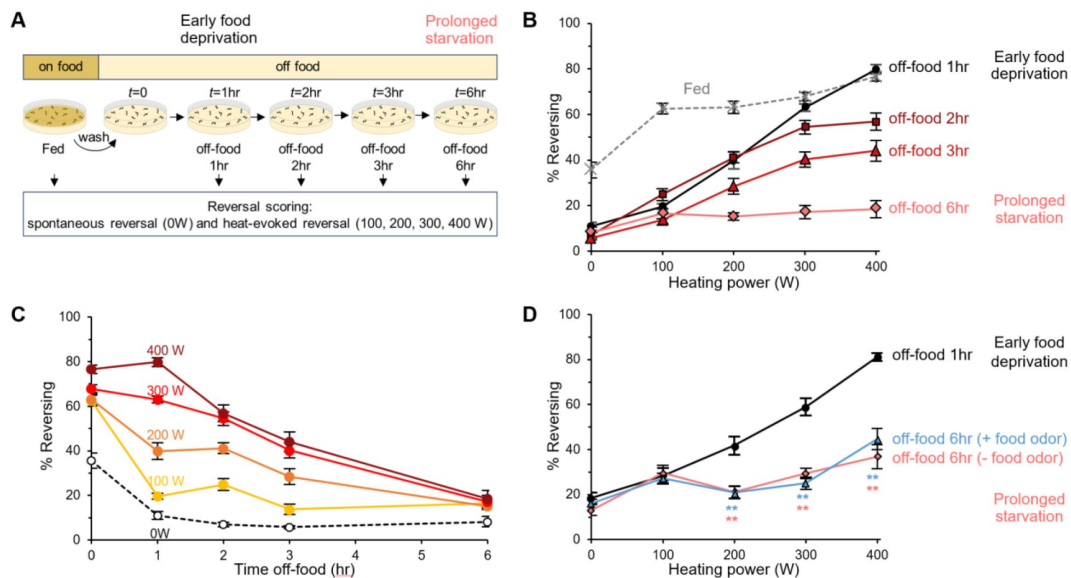


Figure 1

Starvation-dependent thermonociceptive plasticity in *C. elegans*.

A. Schematic of the experimental procedure to quantify the impact of food deprivation on thermonociception in *C. elegans* adult hermaphrodites. Spontaneous and heat-evoked reversals were quantified in fed animals on food (Fed) or off food after 1, 2, 3, and 6 hr of food deprivation. Spontaneous reversals were measured as baseline reversal rate prior to any stimuli (0 W) and heat-evoked reversals were measured during a series of 4-s heat pulses at 100, 200, 300 and 400 W, respectively, delivered with an interstimulus interval of 20 s. **B–C.** Impact of food deprivation on thermonociceptive response, showing progressive attenuation of heat-evoked reversal response in the course of a 6-hr experiment. Results as average fraction of reversing animals (%) quantified in $N \geq 6$ assays, each scoring at least 50 worms. Error bars: S.E.M. The same dataset is presented as heat dose-response curves (B) and as time-course of the heat-evoked response decrease at each heating level (C). Spontaneous reversal rate corresponds to the baseline reversal response in the absence of heating stimuli (0 W). **D.** Effect of prolonged starvation (off-food for 6 hr) in the presence or absence of food odor during the food-deprivation period, indicating that external food odor cues cannot prevent the response reduction. **, $p > .01$ versus the early food deprivation condition (off-food 1 hr), by Holm-Bonferroni post-hoc tests.

neurons induced an increase in spontaneous reversals, but only a mild reduction in heat-evoked responses and left starvation-dependent plasticity largely intact, suggesting a minor contribution of these neurons (**Figure 2 Supplement 1** [↗](#)). By contrast, ablation of AWC or ASI neurons had dramatic effects. Removal of AWC nearly abolished heat-evoked reversal behavior across all stimulus intensities and timepoints (**Figure 2B and D** [↗](#)), confirming its essential role in mediating the thermonociceptive response. Notably, in AWC-ablated animals, the response level was unaffected by starvation, suggesting that AWC might also be required for the expression of starvation-dependent plasticity. In contrast, ASI ablation had almost no effect under early food deprivation but almost entirely suppressed the starvation-induced reduction in heat-evoked reversals observed after prolonged starvation (**Figure 2C and D** [↗](#)). This demonstrates that ASI is specifically required for starvation-dependent modulation of thermonociceptive behavior.

Together, these data show that, under our experimental conditions, AWC is the primary driver of heat-evoked reversals upon early food deprivation, while ASI neurons selectively mediate their suppression under prolonged starvation.

AWC neurons mediate heat-evoked reversal via both glutamate and FLP-6 neuropeptides

We next sought to define the molecular mechanisms by which AWC neurons mediate robust heat-evoked reversals in the early food deprivation condition. AWC neurons are glutamatergic and express several neuropeptides, including FLP-6, INS-22 and NLP-5 [32 [↗](#), 33 [↗](#)]. *eat-4* mutants lacking the vesicular glutamate transporter EAT-4 exhibited strong impairments in heat-evoked responses at high stimulus intensities (300–400 W) and a reduction in spontaneous reversal rates (**Figure 3A** [↗](#)). *flp-6* mutants displayed a marked deficit across a broad, 200–400 W stimulus range, while the impairment in *ins-22* and *nlp-5* mutants was less pronounced and more selectively affected the 200–300 W range of stimulus intensities (**Figure 3C** [↗](#) and **Figure 3-Supplement 1** [↗](#)). This indicates that different communication molecules are used over different thermal ranges to mediate heat-evoked reversals, which is in line with previous genetic analyses having recurrently shown that multiple genetic pathways are recruited for different levels of heat [6 [↗](#), 34 [↗](#)].

Next, we focused on *eat-4* and *flp-6* mutants, showing the strongest phenotype, and carried out cell-specific rescue to demonstrate that AWC neurons represent a relevant source of glutamate and FLP-6 neuropeptides promoting heat-evoked reversals upon early food deprivation (off-food 1hr). Transgenic rescue experiments using either [*AWC^{OFF}::eat-4*] or [*AWC^{ON}::eat-4*] extrachromosomal array containing-lines showed that restoring *eat-4* expression in either AWC^{ON} or AWC^{OFF} alone was sufficient to recover heat-evoked responses to near wild-type levels (**Figure 3B** [↗](#)). Interestingly, [*AWC^{OFF}::eat-4*] transgene not only restored responsiveness but also enhanced reversals at 100 W, potentially reflecting overexpression effects or altered signaling balance with other glutamatergic neurons. Additionally, spontaneous reversal rates were restored only by [*AWC^{OFF}::eat-4*]. A similar rescue approach for *flp-6* expression in AWC revealed that only [*AWC^{OFF}::flp-6*], not [*AWC^{ON}::flp-6*], could rescue heat-evoked behavior, suggesting asymmetry in neuropeptidergic signaling (**Figure 3D** [↗](#)).

Our data indicate that any one of the two AWCs might be sufficient to mediate a large part of the glutamate-dependent reversal response, but that AWC^{OFF} and AWC^{ON} might display some functional asymmetries, regarding the role of neuropeptides and the regulation of spontaneous reversal. To further address the role of AWC^{ON/OFF} asymmetry, we examined *nsy-1* mutants, developing with two AWC^{ON} and *nsy-7*, developing with two AWC^{OFF} [35 [↗](#), 36 [↗](#)]. Both mutants exhibited reduced responses at high heat intensities, with *nsy-1* also showing increased spontaneous reversals (**Figure 3-Supplement 2** [↗](#)).

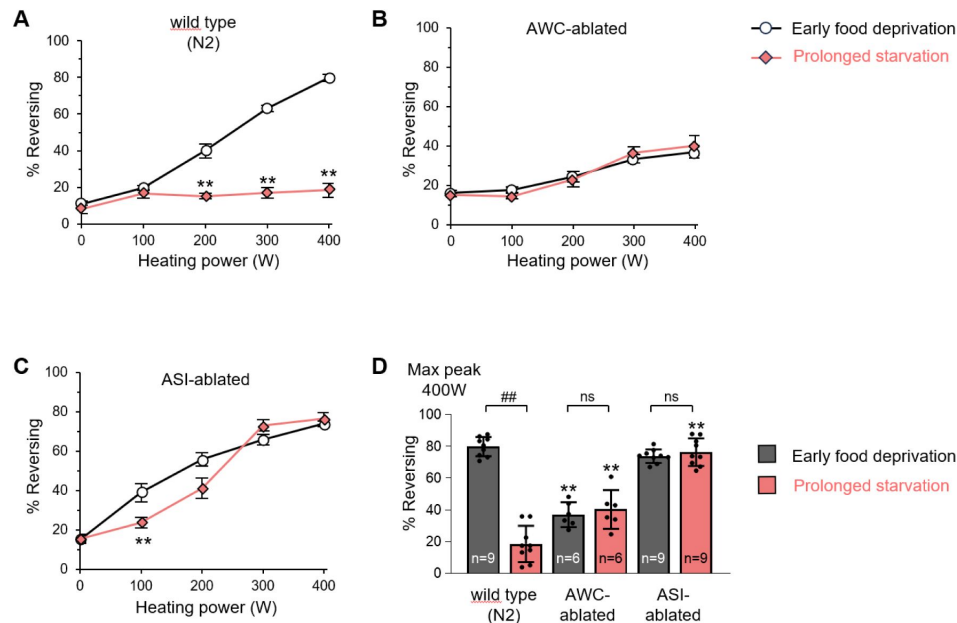


Figure 2

AWC mediates heat-evoked reversals upon early food deprivation and ASI mediates thermonociceptive plasticity upon starvation.

Impact of the genetic ablation of AWC and ASI sensory neurons on thermonociceptive response and starvation-dependent plasticity. **A-C** Comparison of heat-evoked reversals after early food deprivation (off-food 1 hr) and prolonged starvation (off-food 6 hr) in wild type and in transgenic animals with caspase-mediated ablation of indicated neurons. Results are presented as average $\pm$ S.E.M. *, $p < .05$, and **, $p < .01$ versus corresponding heat level in the early food deprivation condition, by Bonferroni post-hoc tests. **D** Comparison for the 400 W heating level across the three genotypes presented in panel A to C. Bars as average, dots as individual assay scores, and error bars as S.E.M. ##, $p < .01$ between early food deprivation and prolonged starvation for each genotype; **, $p < .01$ versus wild type (N2) at the corresponding time point, by Holm Bonferroni post-hoc tests. The total number of assays analyzed per condition, each scoring at least 50 worms, are indicated in panel D.

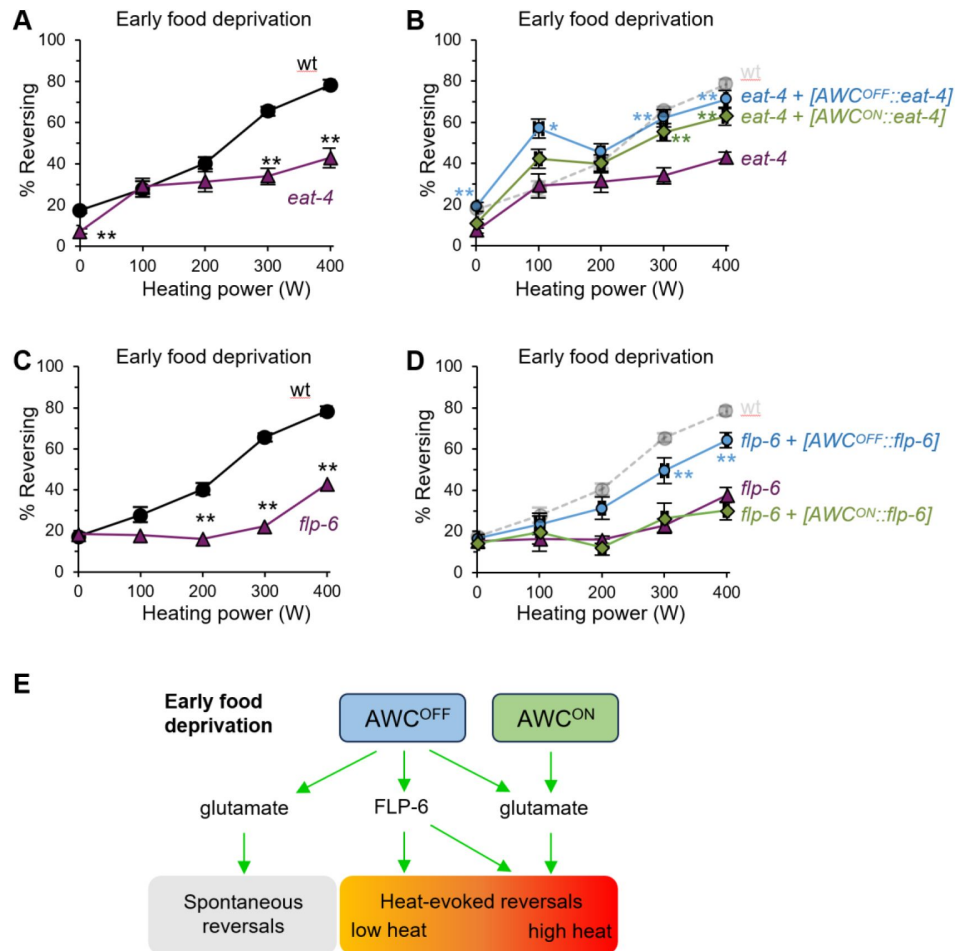


Figure 3

Differential engagement of glutamate and FLP-6 neuropeptides by AWC^{ON} and AWC^{OFF} in the control of spontaneous and heat-evoked reversal at various heat levels.

A-D Comparison of spontaneous (0 W) and heat-evoked reversals (100 to 400 W) after early food deprivation (off-food 1hr) in wild type, *eat-4(ky5)*, *flp-6(ok3056)* and in transgenic animals with AWC subtype-specific rescue of *eat-4* and *flp-6*, respectively. Results are presented as average $\pm$ S.E.M. **, $p < .01$ versus wild type (A and C) and versus non-transgenic mutants (B and D) at respective heat levels, by Holm-Bonferroni post-hoc tests. The number of assays (N), each scoring at least 50 worms, were: wild type, N=15; *eat-4*, N=6; *flp-6*, N=14; *eat-4*+ [AWC^{OFF}::*eat-4*], N=6; *eat-4*+ [AWC^{ON}::*eat-4*], N=9; *flp-6*+ [AWC^{OFF}::*flp-6*], N=7; *flp-6*+ [AWC^{ON}::*flp-6*], N=8. **E** Visual model illustrating the specific contribution of AWC^{OFF} and AWC^{ON} to the regulation of spontaneous reversal, low heat-evoked reversal and high heat-evoked reversal, via the distributed action of glutamate and FLP-6 neuropeptide.

These results indicate that AWC neurons mediate thermosensory responses through both glutamatergic and FLP-6-dependent pathways, each covering distinct heat intensity ranges (**Figure 3E**). While either AWC subtype is sufficient for glutamatergic signaling, FLP-6 neuropeptide function appears more dependent on AWC^{OFF} identity.

Starvation alters the nature of AWC heat responses from deterministic to stochastic

To investigate if and how starvation modulates AWC activity at the cellular level, we recorded heat-evoked calcium dynamics in AWC^{ON} and AWC^{OFF} neurons using YC2.3 cameleon-expressing transgenic lines. We used a microfluidic setup to deliver defined thermal stimuli. From a baseline at 20°C, four stimuli reaching 22, 24, 26 and 28°C, respectively, were sequentially delivered, corresponding approximately to the temperatures reached in the heat-evoked behavioral analysis (**Figure 3C**). We examined responses under both early food deprivation and prolonged starvation conditions. In **Figure 4A and B**, we report average traces and individual traces as heat maps. Individual traces are essential to visualize the stochastic component of the response in some neurons and we clustered traces as ‘calcium up’ (traces showing consistent up-regulation after 24, 26 and 28° stimuli), ‘variable’ (traces showing a mix of up and down regulation) and ‘calcium down’ (traces showing consistent down-regulation after 24, 26 and 28° stimuli). The ‘calcium down’ traces were often, but not always, followed by a ‘rebound’ upregulation.

Upon early food-deprivation, both AWC neuron subtypes responded with a stereotyped, intensity-dependent calcium increase (“calcium up”) in most animals (45/56, 80% for AWC^{OFF} and 37/49, 77% for AWC^{ON}). The remaining traces showed either mixed (“variable”) or “calcium down” profiles.

After prolonged starvation, this response pattern shifted dramatically: the majority of traces became variable or showed consistent calcium down-regulation (17/23, 74% for AWC^{OFF} and 14/17, 82% for AWC^{ON}, both changes being statistically significant by Fisher’s exact test: $p=1.3E-5$ and $p=7.6E-5$, respectively). This shift was reflected in the average traces (**Figure 4A and B**) as a decreased peak amplitude and rightward shift. But the modification of the average trace was largely caused by the redistribution in response types: the dynamic of individual *calcium up* and *calcium down* traces remaining similar in each condition (**Figure 4C and D**). Therefore, AWC neurons, which under early food deprivation respond consistently to heat, enter a heterogeneous and less predictable response mode under prolonged starvation.

In summary, starvation does not simply attenuate AWC calcium responses; it fundamentally alters their dynamics, shifting from a largely deterministic excitation profile to a heterogeneous regime characterized by decreased and variable responses. We speculate that this transformation in thermal encoding might underlie the behavioral plasticity observed during starvation.

ASI is required to shift AWC activity from deterministic to stochastic modes upon starvation

Because ASI neurons are required for the starvation-dependent thermosensory plasticity, dispensable for noxious heat-evoked reversal response, and have been previously shown to signal the feeding state of the animal [37], we hypothesized that ASI neurons might play a modulatory role to mediate the starvation impact on AWC activity patterns. To test this hypothesis, we recorded AWC^{ON} and AWC^{OFF} activity in ASI-ablated animals, which we found to lack starvation-evoked plasticity (**Figure 2C**). In the early food-deprivation condition, 1hr off-food, ASI ablation produced no noticeable difference in AWC calcium activities (compare **Figure 4** with 5). Strikingly different from the situation in animals with intact ASI (**Figure 4**), starvation after 6hr

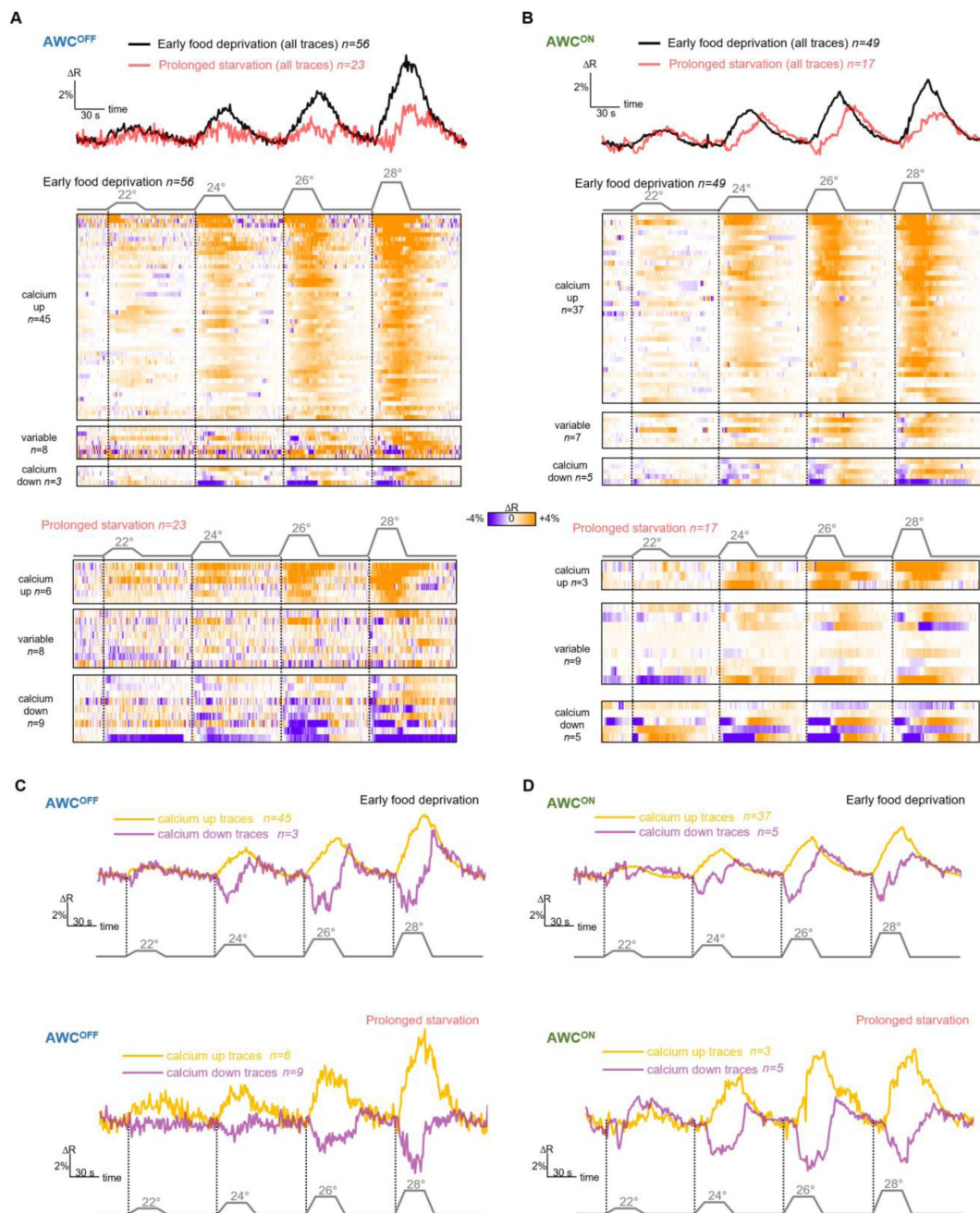


Figure 4

Starvation reconfigures AWC heat-evoked response from a mostly deterministic to a stochastic activity mode.

A-B Calcium activity in AWC^{OFF} (left) and AWC^{ON} (right) in response to a series of four thermal up-steps. Upper plots show the average traces, lower heat maps show individual neuron traces, in the early food deprivation (off-food 1hr) and prolonged starved (off-food 6hr) conditions. **C-D** From the same dataset as in A-B, average traces showing the dynamics of calcium up or calcium down response types with similar amplitude and globally comparable shapes.

off-food produced no noticeable effect in ASI-ablated animals, neither for AWC^{ON}, nor for AWC^{OFF} (Figure 5B). We conclude that the starvation-dependent shift in AWC activity from a deterministic to stochastic pattern is mediated by the neuromodulatory neurons ASI.

Starvation-evoked thermosensory plasticity is promoted by ASI-expressed neuropeptides, including NLP-18 and INS-32, and modulated by glutamatergic signaling

To elucidate the molecular signaling pathway underlying starvation-evoked thermosensory plasticity involving the AWC and ASI sensory neurons, we investigated the role of candidate neurotransmitters expressed in AWC and ASI, respectively.

First, we tested whether starvation-dependent plasticity was preserved in *eat-4* and *flp-6* mutant backgrounds. Even if the heat-evoked response upon early food-deprivation was reduced relative to wild type in *flp-6* mutants, a significant further decline was seen after prolonged starvation (Figure 6B). These results indicate that starvation-dependent plasticity can operate independently of FLP-6. In contrast, *eat-4* mutants displayed markedly elevated heat-evoked responses after prolonged starvation, even exceeding the response level seen in the early food deprivation condition (Figure 6C). This potentiated response could not be rescued by expressing *eat-4* selectively in either AWC^{OFF} or AWC^{ON} neurons (Figure 6D). Interestingly, AWC^{OFF}-specific rescue produced a further potentiation of heat-evoked reversal response to high heat stimuli (Figure 6D, 300 and 400 W), aggravating the phenotype of *eat-4* mutants.

These results are consistent with a model in which glutamatergic signaling regulates heat-evoked reversals in starved animals via two bidirectional drives (Figure 6E). On the one hand, glutamatergic signalling—originating from AWC^{OFF}—up-regulates reversals in response to high heat stimuli, thus contributing to prevent starvation-induced thermosensory plasticity. On the other hand, glutamatergic signaling—originating from neurons other than AWC—down-regulates reversals over a broad range of heat intensities, thus promoting starvation-induced thermosensory plasticity. The latter glutamatergic signaling effect seems to be more dominant.

Second, since ASI is required for starvation-evoked plasticity but does not release classical neurotransmitters, we hypothesized that its function is mediated by neuropeptidergic signaling. We therefore screened for defects in starvation-dependent plasticity in mutants for neuropeptide genes known to be expressed in ASI: *ins-4*, *ins-6*, *ins-32*, and *nlp-18* [32, 33, 38, 39]. *ins-4* mutants exhibited normal plasticity, with robust responsiveness under early food deprivation and significantly reduced responses after prolonged starvation (Figure 7A). Mutants for *ins-6*, *ins-32*, and *nlp-18* all showed reduced responsiveness in the early food deprivation condition, but abnormally elevated responses after prolonged starvation (Figure 7A). This phenotype was qualitatively similar to the potentiation phenotype of *eat-4* mutants (Figure 6C). We performed further analyses with *ins-32* and *nlp-18* mutants, since they produced the most significant potentiation over a large range of heat intensities. To confirm whether ASI is a relevant source of INS-32 and NLP-18 neuropeptides, we conducted ASI-specific rescue experiments in *ins-32* and *nlp-18* mutants. In *ins-32* mutant background, ASI-specific expression of *ins-32* produced a strong rescue effect, significantly restoring reversal response in early food-deprive animals (Figure 7B) and significantly reducing heat-evoked reversal response under the starvation condition (Figure 7C). In *nlp-18* mutant background ASI-specific expression of *nlp-18* did not restore normal responsiveness after 1 hr off-food, but partially restored the response decrease after 6 hr of starvation. These results indicate that ASI is a functionally relevant source of both INS-32 and NLP-18 to promote starvation-evoked thermosensory inhibition, although they do not exclude the possibility of contributions from other sources (Figure 7F). Furthermore, INS-32 from ASI and NLP-18 from additional neurons might promote reversals upon early food deprivation.

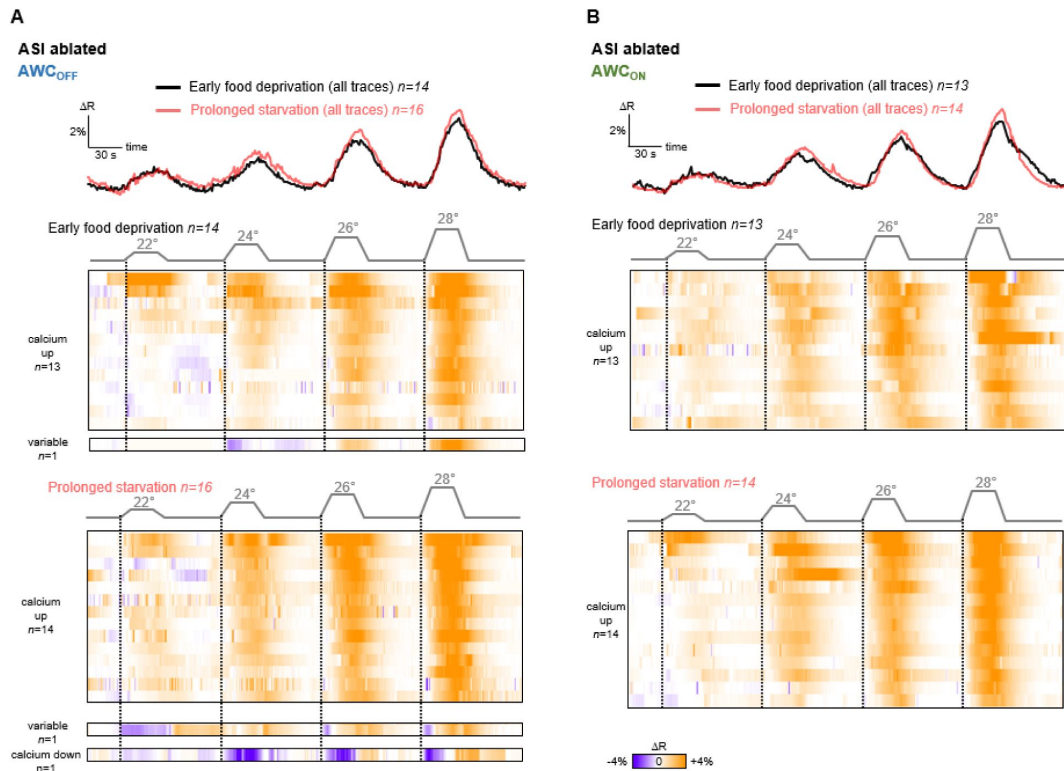


Figure 5

ASI ablation prevents AWC activity pattern reconfiguration upon starvation.

(A-B) Calcium activity in AWC^{OFF} (A) and AWC^{ON} (B) in response to a series of four thermal up-steps in ASI-ablated animals. Upper plots show the average traces, lower heat maps show individual neuron traces, in the early food deprivation (off-food 1hr) and prolonged starved (off-food 6hr) conditions. The starvation impact seen in animals with intact ASI ([Figure 4](#)) is absent when ASI is ablated.

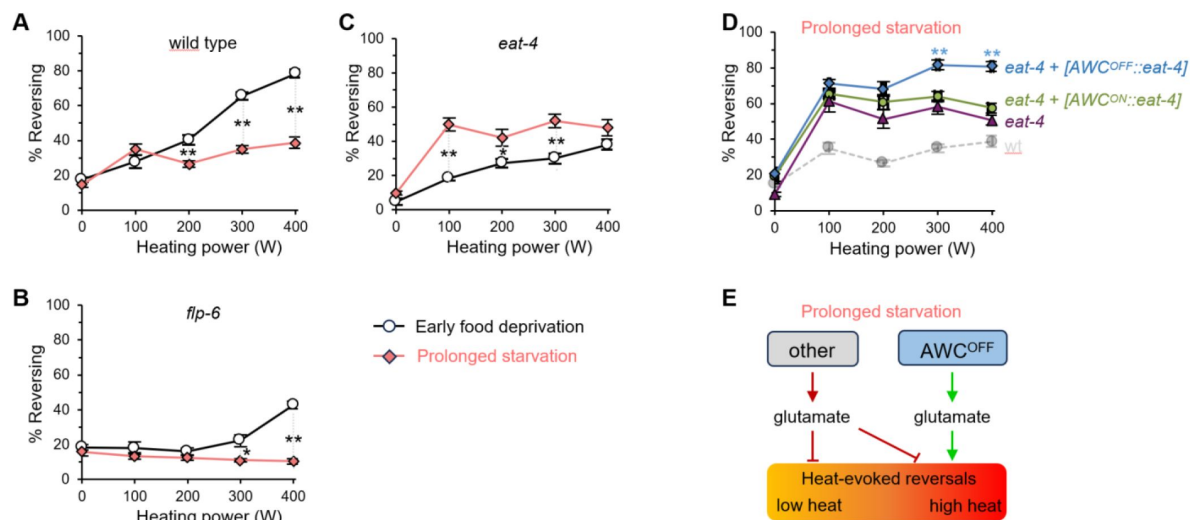


Figure 6

Bidirectional glutamate signaling actions modulate heat-evoked reversals following starvation.

A–C Comparison of heat-evoked reversals after early food deprivation (off-food 1hr) and prolonged starvation (off-food 6hr) in wild type, *eat-4(ky5)*, *flp-6(ok3056)*. **D** Analysis of transgenic animals with AWC subtype-specific rescue of *eat-4* after prolonged starvation (off-food 6hr). Results are presented as average $\pm$ S.E.M. **, $p < 0.01$ between the two food-deprivation time points (A–C) and versus non-transgenic mutants (D) at respective heat levels, by Holm-Bonferroni post-hoc tests. The number of assays (N), each scoring at least 50 worms, were: wild type, $N=15$; *eat-4*, $N=6$; *flp-6*, $N=14$; *eat-4*+*[AWC^{OFF}::eat-4]*, $N=6$; *eat-4*+*[AWC^{ON}::eat-4]*, $N=9$. **E** Visual model illustrating the bidirectional effect of glutamatergic signaling from AWC^{OFF} and from unidentified non-AWC neurons (other). Red: reversal-suppressing pathway; Green: reversal-promoting pathway.

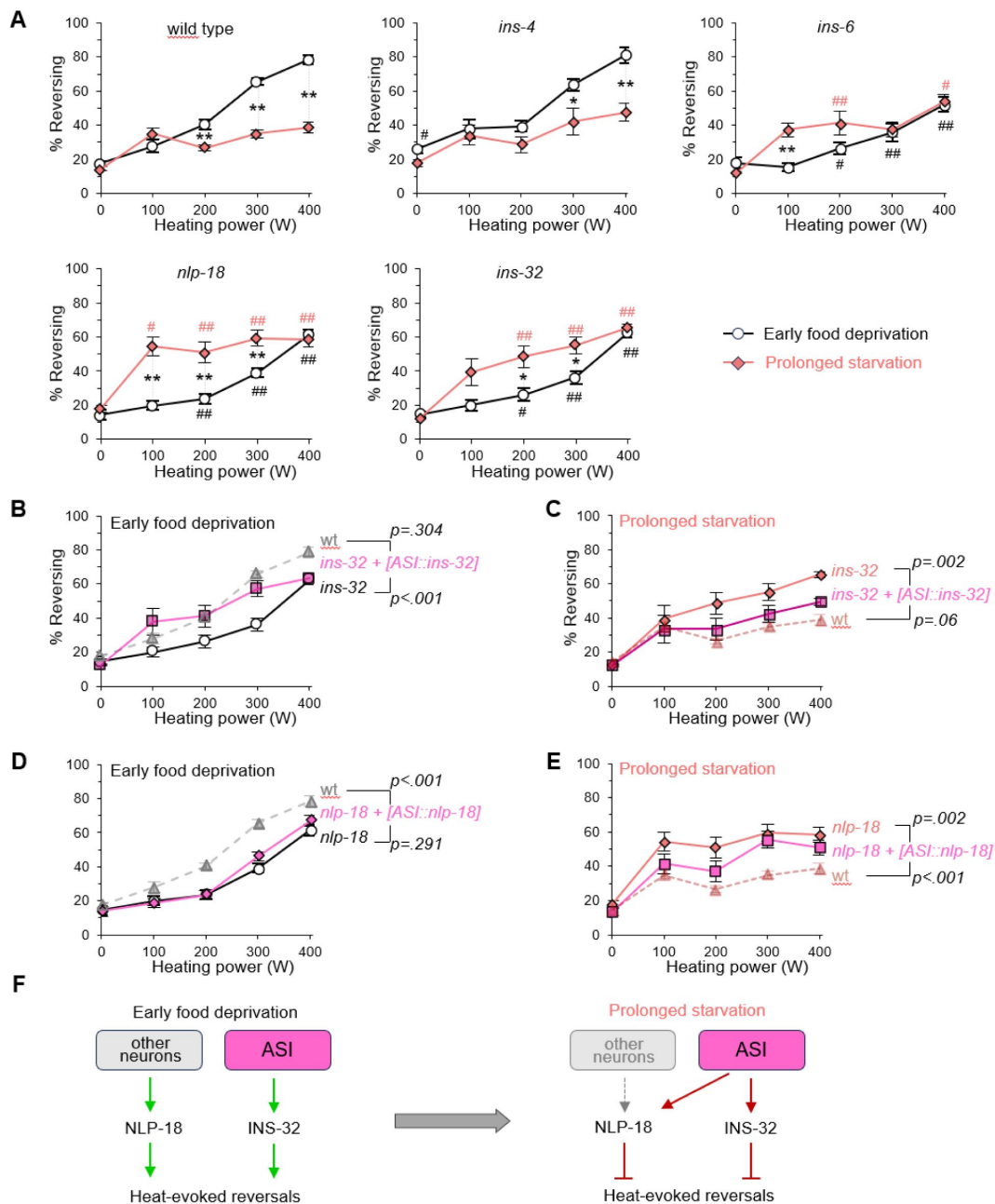


Figure 7

Starvation reconfigures neuropeptidergic signaling by ASI.

A Comparison of heat-evoked reversals after early food deprivation (off-food 1hr) and prolonged starvation (off-food 6hr) in wild type, *ins-4(ok3534)*, *ins-6(tm2008)*, *nlp-18(ok1557)* and *ins-32(tm6109)*. *, $p < 0.05$, and **, $p < 0.01$ between the two food deprivation time points, by Holm Bonferroni post-hoc tests. #, $p < 0.05$, and ##, $p < 0.01$ versus wild type (N2) at the corresponding time point, by Holm Bonferroni post-hoc tests. **B-E** Impact of transgenic rescue in *ins-32* (B-C) and *nlp-18* (D-E) background. Two-way ANOVA showing no significant heating power $\times$ genotype interaction, but a significant genotype main effect, post-hocs tests were conducted on the genotype factor only. Indicated p values were corrected with Holm Bonferroni correction. **F** Model of the bidirectional actions and the sources for NLP-18 and INS-32 neuropeptides in the modulation of heat-evoked reversal. Red: reversal-suppressing pathway; Green: reversal-promoting pathway.

Collectively, our results (Figure 6 and 7) show that starvation-dependent thermonociceptive plasticity is orchestrated by multiple opposing signaling pathways in which neuropeptides and glutamate signals from different neurons produce context-dependent, bidirectional effects.

Discussion

Our study reveals that starvation profoundly modulates thermonociceptive behavior in *C. elegans* by engaging distinct neurons, altering neural response dynamics, and mobilizing opposing neuromodulatory pathways. This behavioral plasticity, manifesting as a progressive attenuation of heat-evoked reversals under prolonged food deprivation, depends on a shift in the internal state that is not rescued by food-associated olfactory cues. By dissecting the underlying circuits and signaling mechanisms, we uncover two key regulatory nodes involving AWC and ASI neurons, which work in order to integrate thermal and nutritional cues into an adaptive behavioral output, as depicted in the visual model in Figure 8.

AWC neurons are central drivers of heat-evoked reversal behavior via glutamate and neuropeptides

Our findings confirm that AWC sensory neurons are critical mediators of thermonociceptive behavior. Previous experiments carried out with extreme heating powers (10 °C raise within tens of milliseconds, [21]) had shown differential responsiveness in the two AWCs subtypes and a large behavioral impact was found in mutants affecting AWC asymmetries. The heating intensity range used in the present study (~0.5 to 2°C per second) triggered a similar withdrawal response, which almost entirely relied on the presence of intact AWC neurons. However, under our experimental conditions, we did not find major differences between AWC^{ON} or AWC^{OFF} subtypes in terms of calcium dynamics, nor did mutations affecting their asymmetry produce radical effects. Therefore, noxious heat-evoked activity in AWC varies widely according to context, which is in line with previous literature [18, 20, 25]. Interestingly, the role of AWC is distinct from that of AFD and FLP neurons, which are canonically linked to thermosensation and nociception [5, 14, 40, 41], but contribute only modestly to heat-evoked behavior in our assays conditions with between 1 and 6 hrs of food deprivation.

We further demonstrate that AWC neurons use both glutamatergic and neuropeptidergic signals to encode heat intensity, with glutamate (via EAT-4) primarily driving responses to stronger stimuli and the FLP-6 neuropeptide acting over a broader thermal range. Functional asymmetry between AWC^{ON} and AWC^{OFF} emerges in our rescue experiments, with AWC^{OFF} more effective in restoring heat-evoked behavior, particularly through FLP-6. This suggests that AWC^{OFF} may have privileged access to neuropeptidergic regulatory machinery or downstream targets, although glutamatergic signaling appears largely redundant across AWC subtypes. Together with previous findings, our results refine our appreciation of how AWC responds to heat and how developmental asymmetries might modulate aversive response to thermal cues over a very broad intensity range.

Starvation reshapes AWC responses from deterministic to stochastic

A key observation of this work is that prolonged starvation transforms heat-evoked AWC calcium responses from a stereotyped, intensity-dependent activation pattern to a heterogeneous, less predictable regime. This shift is not merely a quantitative attenuation but a qualitative reorganization of response states, including frequent calcium downregulation or variable traces, which in the same animal can comprise both calcium up- and down-regulation of calcium levels during a single stimulus train. We speculate that this transition from deterministic to stochastic response profiles reflects a fundamental reconfiguration of sensory processing under prolonged

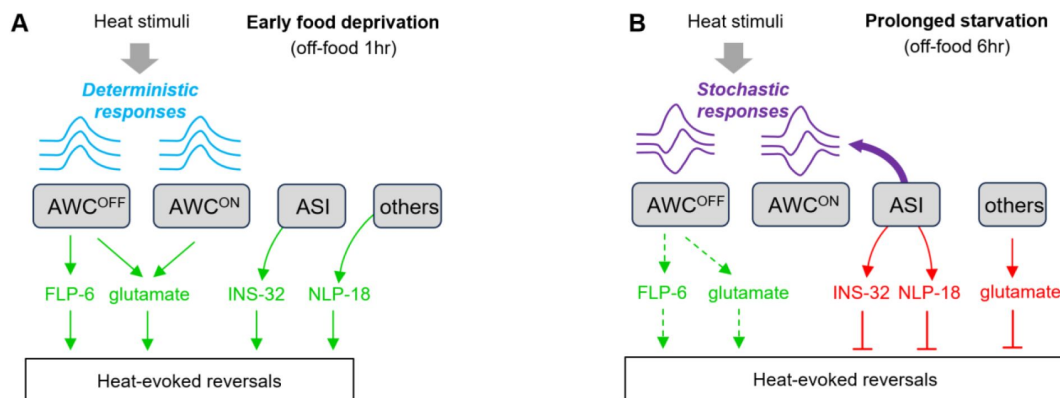


Figure 8

Visual model of the cellular and molecular signaling pathways mediating and modulating heat-evoked reversals according to feeding state.

A Situation upon early food deprivation (off-food 1hr). The two AWC neurons produce mostly deterministic calcium elevations in response to heat stimuli and make a major contribution to heat-evoked reversals. Glutamate signaling from AWC^{ON} and AWC^{OFF}, as well as FLP-6 neuropeptide signaling from AWC^{OFF} mediates heat-evoked reversals. INS-32 neuropeptide from ASI neurons, as well as NLP-18 neuropeptides from an unidentified source, also promote heat-evoked reversals. **B** Situation following prolonged starvation (off-food 6hr). Starvation causes several functional changes. First, AWC calcium activity pattern switches from a deterministic mode to a stochastic mode, with a mix of calcium elevations, calcium decreases and variable responses; this effect is mediated by ASI. Second, INS-32 and NLP-18 neuropeptides switch from a reversal-promoting effect to a reversal-suppressing effect; for INS-32, ASI remains a relevant peptide source, whereas for NLP-18 ASI becomes a relevant peptide source. Third, glutamatergic signaling switches from a mostly reversal-promoting effect to a reversal-suppressing effect; this switch is associated with different glutamate sources (in AWC and non-AWC neurons, respectively). Of note, residual reversal response upon starvation might be mediated by FLP-6 and glutamate from AWC^{OFF}. In summary, our findings highlight a complex reconfiguration of the thermoresponsive circuit following starvation, which includes thermosensory encoding change and complex neuromodulation using bidirectional signaling molecules that produce reversal-promoting (green pathways) or reversal-suppressing effects (red pathways) in a context-dependent manner.

nutrient deprivation. In an early food-deprived state, AWC neurons may signal heat intensity in a consistent and reliable manner to promote aversion. Under prolonged starvation, however, dampened and variable AWC responses may bias the animal away from escape behaviors, thus prioritizing food-seeking over threat avoidance. This state-dependent reconfiguration may serve as a flexible strategy to balance risk avoidance and metabolic needs, as elaborated below.

ASI neurons modulate AWC dynamics and promote plasticity via neuropeptides

Our data demonstrate that ASI neurons are dispensable for baseline thermnociceptive responses but essential for starvation-induced behavioral plasticity. Ablation of ASI fully abolished the starvation-dependent attenuation of heat-evoked reversals and prevented the stochastic shift in AWC calcium dynamics. These results suggest that ASI gates the modulation of AWC activity based on internal nutritional state. ASI might also modulate the circuit downstream of AWC. Interestingly, in two previous studies, starvation-evoked down-regulation of AWC-dependent thermotactic behavior was intact when ASI was either transiently or chronically inhibited [39, 42]. Therefore, starvation most likely uses a different circuitry to modulate noxious-heat evoked reversal than for thermotaxis modulation.

We further identify two ASI-expressed neuropeptides, NLP-18 and INS-32, as relevant mediators of this plasticity. Mutants for the corresponding pro-neuropeptide genes displayed abnormal behavioral adaptation to starvation, and transgenic expression of either peptide in ASI partially rescues the plasticity defect. ASI was previously shown to mediate starvation-evoked phenotypic plasticity regarding developmental trajectories—via TGFβ and the insulin-like peptide DAF-28 [43, 44]—, food-seeking exploration in starved larvae - via NLP-24 opioid-like neuropeptide [45], the curvature taken during head-touch evoked escape response, via NLP-18 [46], and ASH-mediated octanol avoidance response via multiple NLP neuropeptides [47]. Our findings now add context-dependent aversive thermosensory behavior regulation to the growing list of satiety signaling and stress integration-related functions regulated by ASI neurons. It remains to be determined how ASI integrates systemic cues of metabolic status and how its neuropeptidergic outputs interact with AWC-neural pathway at the synaptic or circuit level to regulate noxious heat avoidance.

Opposing neuromodulatory influences shape behavioral flexibility in order to balance protection and exploration strategies

The balance between starvation-induced suppression and preservation of thermnociceptive behavior appears to be controlled by a set of signaling molecules, whose effect can be bidirectional, operating in a context-dependent manner. Upon early food deprivation, neuropeptides, including INS-32 released from ASI and NLP-18 from an unidentified source, as well as glutamatergic signals from AWC neurons promote thermnociceptive behaviors (**Figure 8A**). Upon prolonged starvation, the same signaling molecules can produce the opposite effect and suppress this behavior (**Figure 8B**). For NLP-18 and glutamate the differential effect is associated with a potentially different source, e.g., non-AWC neurons for glutamate. For INS-32, our cell-specific rescue experiment suggests that ASI is a functionally relevant source in the two contexts. However, we cannot rule out the implication of additional neurons, nor an overexpression effect. While additional studies will be needed to decipher the context-dependent mode of action of each of these signaling molecules, our findings suggest a regulatory architecture in which bidirectional neuromodulatory signals integrate metabolic context with sensory drive to determine behavioral priorities. Rather than globally silencing the thermnociceptive pathway, starvation might well reconfigure it through a layered combination of neuronal state shifts, circuit-level inhibition, and peptide-based tuning. This complexity may afford the animal greater flexibility in selecting context-appropriate responses under competing environmental and internal pressures. From an ecological and evolutionary perspective, the downregulation of nociceptive

responses during prolonged starvation likely reflects a strategic shift in behavioral priorities. In the face of acute noxious stimuli, animals typically favor rapid escape to minimize injury—a protective response. However, under prolonged food deprivation, such protection may come at the cost of missed opportunities for locating food. In this context, a heightened aversive response could lead to excessive avoidance of mildly threatening environments that may nevertheless contain nutritional resources. In addition, because worm exploration strategies rely on modulating locomotion speed, the duration of forward locomotion bouts and the frequency of re-orientation events, often associated with reversals, the down-regulation of reversal response is directly relevant to the rate of animal dispersion in the environment [48, 49]. By reducing the sensitivity or reliability of nociceptive pathways— particularly through modulating AWC output— *C. elegans* may adopt a more exploratory behavioral mode, tolerating greater risk to increase the chances of finding food. This trade-off between protection and exploration mirrors similar state-dependent shifts observed in other species [50, 51] suggesting a conserved strategy wherein internal metabolic needs dynamically reshape threat evaluation and behavioral responses.

Methods

C. elegans maintenance and strains list

C. elegans strains were maintained according to standard techniques on nematode growth medium (NGM) agar plates seeded with OP50. Animal synchronization was made by treating gravid adults with standard hypochlorite-based procedure. File S1 includes a list of strains used in the present study.

Cloning and transgenesis

DNA prepared with a GenElute HP Plasmid miniprep kit (Sigma) was microinjected in the gonad to generate transgenic lines according to a standard protocol [52]. Promoter-containing Entry plasmids (Multi-site Gateway slot 1) were constructed by PCR using N2 genomic DNA as template and primers flanked by attB4 and attB1r recombination sites; the PCR product being cloned into pDONR-P4-P1R vector (Invitrogen) by BP recombination. Coding sequence-containing Entry plasmids (Multi-site Gateway slot 2) were constructed by PCR using N2 cDNA as template and primers flanked by attB1 and attB2 recombination sites; the PCR product being cloned into pDONR_221 vector (Invitrogen) by BP recombination. Expression plasmids for transgenesis were created through LR recombination reactions (Gateway LR Clonase, Invitrogen) as per the manufacturer's instructions. DNA prepared with a GenElute HP Plasmid miniprep kit (Sigma) was microinjected in the gonad to generate transgenic lines according to a standard protocol (Evans, 2006). File S1 includes a list of plasmids and primers used in this study.

Behavioral assays

Preparation of worms

All the behavioral experiments were performed in young adult animals cultivated at 20°C. For the Fed condition, 80 animals were transferred to a NGM plate fully covered with OP50 bacterial lawn 18 hours before the experiment. For starved condition, fed animals were washed 3 times in 1.5 mL collection tubes with M9 (pre-equilibrated at the growth temperature of worms) to remove OP50. During washes, worms were left to settle to the bottom of the tubes by gravity. About 80 animals were then plated on unseeded NGM plates with a drop of M9 and left to air dry for 5 min. Duration of starvation was considered from the time of the start of the wash. During starvation, animals were kept in the incubator at 20°C.

Heat-stimulation protocol

Heat stimulation was delivered using previously described INFERNO system [30]. The stimulation program consisted of a baseline period of 40 s without any heat stimulation (to determine spontaneous reversal rate), 4 s with 100 W heating (1 IR lamp turned on), 20 s of interstimulus interval (ISI), 4 s with 200 W heating (2 lamps turned on), 20 s of ISI, 4 s with 300 W heating (3 lamps turned on), 20 s of ISI and 4 s with 400 W heating (4 lamps turned on) as previously described. The peak temperatures reached at the surface of the plate were ~22, ~24, ~26 and ~28°C, for each of the four heating powers, respectively.

Movie recording and analysis

During the stimulation program, worm plates were filmed using a DMK 33U×250 camera and movies acquired with the IC capture software (The Imaging Source), at a 1600×1800 pixel resolution, at 8 frames per second, and the resulting AVI file was encoded as Y800 8-bit monochrome. Behavioral recordings were analyzed using the Multi-Worm Tracker 1.3.0 (MWT) [53]. A custom Python script was used to flag reversal events and report the frame during which they occurred. The movie time course was separated into 4-s bins and the fraction of animals reversing in each bin was extracted as the primary output for subsequent analyses.

Calcium imaging

Calcium imaging was performed using a Leica DMI6000B inverted epifluorescence microscope equipped with a HCX PL Fluotar L40x/0.60 CORR dry objective, a Leica DFC360FX CCD camera, an EL6000 light source, and equipped with fast filter wheels for FRET imaging (excitation filter: 427 nm (BP 427/10); emission filters 472 (BP 472/30) and 542 nm (BP 542/27) [14]. Calcium levels in AWC^{ON} and AWC^{OFF} neurons were monitored using cameleon YC2.3 sensor. Animals cultivated at 20°C were glued, maintained at 20°C for 90 s of baseline imaging acquisition and subjected to 30s heat stimulation at 22°C, 24°C, 26°C and 28°C using CherryTemp microfluidic system by Cherry Biotech, with 60s ISI. Imaging YFP/CFP ratio of the YC2.3 cameleon indicator was used to analyze relative changes in response to short-lasting stimuli. The imaging was performed 1 h and 6 h after food deprivation.

Statistical analyses

ANOVAs and post hoc tests, using Bonferroni–Holm correction, were conducted using GraphPad Prism. Replicate numbers are defined in the Figures.

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Figure 2-supplement 1

Heat-evoked reversal upon early food deprivation and starvation-dependent plasticity are largely intact in AFD and FLP-ablated animals.

Impact of the genetic ablation of candidate thermo-responsive neurons on thermonociceptive response and starvation-dependent plasticity. **A-C** Comparison of heat-evoked reversals after early food deprivation (off-food 1hr) and starvation (off-food 6 hr) in wild type and in transgenic animals with caspase-mediated ablation of indicated neurons. Results are presented as average $\pm$ S.E.M. *, $p < .05$, and **, $p < .01$ versus corresponding heat level in the early food deprivation condition, by Bonferroni post-hoc tests. **E** Comparison for the 400 W heating level across the three genotypes presented in panel A to C. Bars as average, dots as individual assay scores, and error bars as S.E.M. ##, $p < .01$ between early food deprivation and prolonged starvation for each genotype; **, $p < .01$ versus wild type (N2) at the corresponding time point, by Holm Bonferroni post-hoc tests. The total number of assays analyzed per condition, each scoring at least 50 worms, are indicated in panel D.

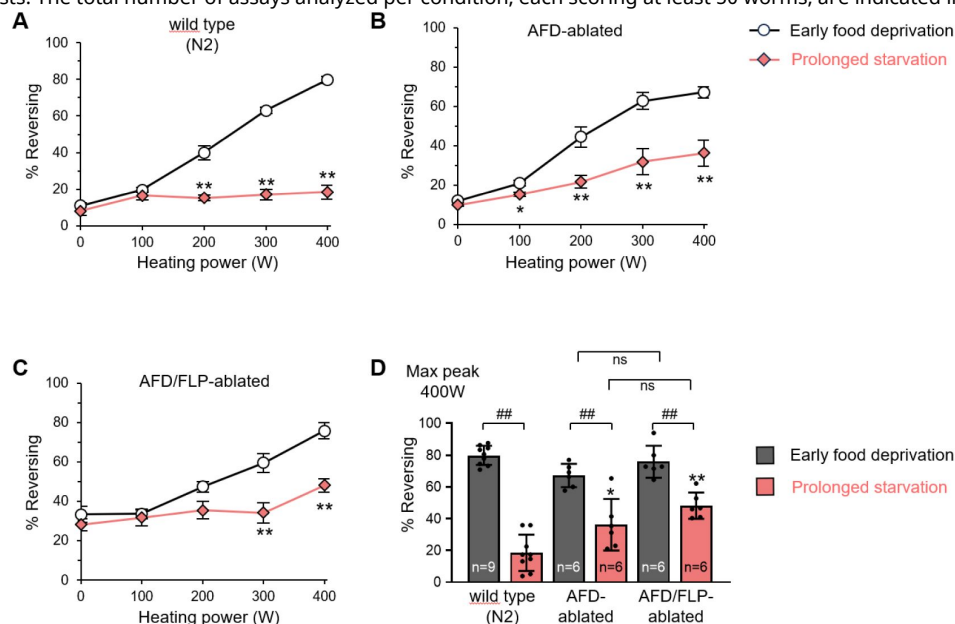
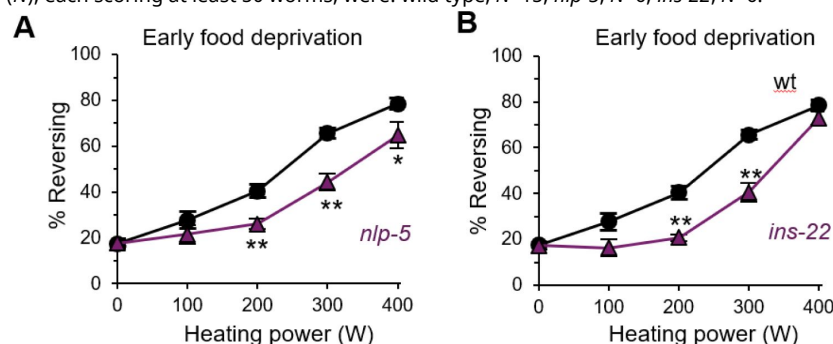


Figure 3-Supplement 1

A-B Comparison of spontaneous (0 W) and heat-evoked reversals (100 to 400 W) after early food deprivation (off-food 1hr) in wild type, *nlp-5(ok1981)* and *ins-22(ok3616)*.

Results are presented as average $\pm$ S.E.M. *, $p < .05$ and **, $p < .01$ versus wild type by Holm-Bonferroni post-hoc tests. The number of assays (N), each scoring at least 50 worms, were: wild type, N=15; *nlp-5*, N=6; *ins-22*, N=6.



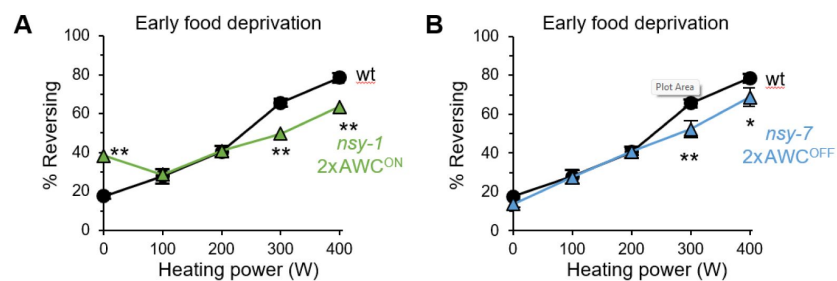


Figure 3-Supplement 2

AWC subtype differentiation is partly required for proper heat-evoked reversal response after 1hr off food.

A-B Comparison of spontaneous (0 W) and heat-evoked reversals (100 to 400 W) after early food deprivation (off-food 1hr) in wild type, *nsy-1(ok593)*, in which both AWCs differentiate as AWC^{ON} and *nsy-7(tm3080)*, in which both AWCs differentiate as AWC^{OFF}. Results are presented as average $\pm$ S.E.M. *, $p < .05$ and **, $p < .01$ versus wild type by Holm-Bonferroni post-hoc tests. The number of assays, each scoring at least 50 worms, were: wild type, $N=15$; *nsy-1*, $N=6$; *nsy-7*, $N=6$.

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Reviewer #1 (Public review):

This study by Thapliyal and Glauser investigates the neural mechanisms that contribute to the progressive suppression of thermonociceptive behavior that is induced under conditions of starvation. Several previous studies have demonstrated that when starved, *C. elegans* alters its preferences for a variety of sensory cues, including CO₂, temperature, and odors, in order to prioritize food seeking over other behavioral drives. The varied mechanisms that underlie the ability of internal states to alter behavioral responses are not fully understood,

however there is growing evidence for a role by neuropeptidergic signaling as well as capacity for functionally distinct microcircuits, formed by distinct internal states, to trigger similar behavior outcomes.

Within the physiological range of *C. elegans* (~15-25°C), starvation triggers a profound reduction in temperature-driven thermotaxis behaviors. This reduction involves the recruitment of the amphid sensory neuron pair AWC. The AWC neurons primarily act to sense appetitive chemosensory cues, however under starvation conditions begin to display temperature responses that previous studies have linked to the reduction in thermotaxis navigation. Here, Thapliyal and Glauser investigate the impact of starvation on thermonociceptive responses, innate escape behaviors that are triggered by exposure to noxious temperatures above 26°C or rapid thermal stimuli below 26°C. They compare the strength of thermonociceptive behaviors, specifically heat-triggered reversals, in worms experiencing either early food deprivation (1 hour off food) or prolonged starvation (6 hours off food). Their experiments demonstrate a progressive loss of heat-triggered reversals that is mediated by AWC and ASI neurons, as well as both glutamatergic and neuropeptidergic signaling.

At the level of neural activity, this study reports that the transition from early food deprivation to prolonged starvation reconfigures the temperature-driven activity of AWC neurons from largely deterministic to stochastic. This finding is interesting in light of previous work that reported the opposite transition (from stochastic to deterministic) in temperature-driven AWC responses when comparing well-fed worms to those kept from food for 3 hours. This study also identifies neural and genetic mechanisms that contribute to differences in thermonociceptive responses at +1 versus +6 hours starvation; confusingly, these mechanisms are partially distinct from those that contribute to differences in negative thermotaxis behaviors in well-fed and +3 hours starvation worms (Takeishi et al 2020). A limitation of this manuscript is that these differences are not particularly acknowledged or addressed, other than the hypothesis that independent mechanisms underlie negative thermotaxis versus thermonociceptive stimuli. However, this suggestion is not experimentally verified. Multiple additional aspects of this study make the results difficult to synthesize with existing knowledge, including 1) differences in - and insufficient discussion of - the magnitude and kinetics of thermal stimuli; 2) this study's use of "heating power" rather than temperature values when presenting behavioral results; 3) the use of +1 hours starvation as a baseline instead of well-fed worms. Indeed, this last point reflects a noticeable experimental result that differs from previous studies, namely that at room temperature the basal movements of well-fed and starved worms are not different. Such a surprisingly result warrants further quantification of worm mobility in general and could have prompted a set of experiments directly testing previously published thermal conditions, to demonstrate that the new effects reported arise specifically from the use of thermonociceptive stimuli, as hypothesized. Finally, a previous report (Yeon et al 2021) demonstrated differences in the impact of chronic versus acute neural silencing on starvation-dependent plasticity in the context of negative thermotaxis. We therefore wonder whether similar developmental compensation impacts the neural circuits that contribute to starvation-dependent plasticity in the thermonociceptive responses.

A weakness of this manuscript is that the introduction is insufficiently scholarly in terms of citations and the description of current knowledge surrounding the impact of internal state on sensory behavior, particularly given previous work on the impact of feeding state on thermosensory behavioral plasticity (Takeishi et al 2020, Yeon et al 2021) and chemosensory valence (Banerjee et al 2023, Rengarajan et al 2019, etc). Similarly, the authors' commanding knowledge of the distinction between thermotaxis navigation (especially negative thermotaxis) and thermonociceptive behaviors could be communicated in more depth and clarity to the readers, in order to contextualize this study's new findings within the previous literature.

Nevertheless, this study represents a solid addition to the growing evidence that *C. elegans* sensory behaviors are strongly impacted by internal states, and that neuropeptidergic signaling plays a key role in mediating behavioral plasticity. To that end, the authors have provided solid evidence of their claims.

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Reviewer #2 (Public review):

In this work Thapliyal and Glauser tried to provide mechanistic understanding by which animals modulate their neural circuit responses to control nociceptive behavior on the basis of the dynamic internal feeding state. It is an important study that adds to growing body of evidences coming from multiple model systems. They have used elegant genetics, behavioral and Ca-imaging experiments to demonstrate how the auxiliary thermosensory neuron pair, AWC and one of the internal state sensing interneuron pair, ASI, respond to dynamic internal starvation-state to modulate behavioral response to noxious heat. Interestingly, these neuron pairs use distinct molecular mechanisms along with some other unidentified neurons to suppress heat-induced reversal response under short-term and prolonged starvations. The experiments are well performed that support most of the claims and provide important framework for future studies.

I have some queries that if answered, will certainly enhance the study,

- (1) The results suggests that ASI is one of the primary drivers for the starvation-evoked behavioral plasticity, which regulates AWC activity under prolonged starvation. It raises many important questions including, a) how starvation modulates ASI response to heat? b) under prolonged starvation, whether ASI also promotes other, non-AWC, glutamatergic inhibitory neurons to suppress heat-induced reversal and how?
- (2) How does ASI regulate AWC activity? In the proposed model (figure 8) authors suggested an independent, unknown signal, other than INS-32 and NLP-18, from ASI to regulate AWC activity. However, from the results the existence of another signal is not very clear.
- (3) Previously, Takeishi et. al., showed that *ins-1* dynamically modulates AWC-AIA mediated thermotaxis behavior based on the feeding state of the animal. It raises questions whether *ins-1* also contributes to noxious heat-induced reversal behavior.
- (4) Experiments with AWC fate conversion mutants (*nsy-1* and *nsy-7*) were very good ideas, however the results obtained were confusing. *flp-6* mutant data suggests *AWCoff* would be essential for heat induced reversal, especially at the low intensity stimulus level. However, *nsy-1* mutant forming two *AWCon* neurons showed complete rescue at the low heat level, which is quite opposite. Similarly, although less prominent, *eat-4* rescue experiments suggested both *nsy-1* and *nsy-7* should behave normally at high heat condition, which was not the result observed.

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Reviewer #3 (Public review):

Summary:

Thapliyal and Glauser show that hunger alters how *C. elegans* respond to noxious thermal stimuli. Using targeted neural ablation, mutant analysis, and live-cell functional imaging the authors demonstrate that hunger changes the properties of AWC sensory neurons, which

sense noxious heat. The authors further show that effects of hunger on nociception require ASI neurons, which are known to respond to hunger and mediate effects of food deprivation on behavior. Finally, the study uses mutant analysis to implicate glutamate and specific neuropeptides in thermal nociception and in modulation of nociceptors by hunger-responsive neurons.

Strengths:

The study clearly shows a strong effect of hunger on nociception and documents a striking effect of hunger on the intrinsic properties of AWC sensory neurons, which respond to noxious heat. The study also clearly and compellingly demonstrates that ablation of hunger-responsive ASI neurons blocks effects of hunger on nociceptive AWCs. These data, which constitute the kernel of the manuscript, are striking and exciting.

Weaknesses:

The study has some weaknesses that the authors should address.

- (1) Ablation of AWC neurons alters the basal sensitivity to noxious heat stimuli. This should be clearly noted in the description of the result and warrants some discussion.
- (2) Throughout the study it seems that data are replotted in multiple figure panels. The authors should clearly indicate in figure legends when this occurs. Also, the authors should ensure that statistical tests requiring multiple comparisons are correctly implemented and reflect the number of times experimental data are compared to a single set of control data.
- (3) How ASIs modulate AWCs remains unclear. The authors find that loss of INS-6, an insulin-like peptide provided by ASIs, partially recapitulates the effect of ASI ablation. This observation is not further developed and instead the authors characterize other secreted factors that seem to mediate sensitization of animals to noxious heat stimuli. While it is interesting that there are multiple opposing inputs into the nociceptor circuit, the essential connection between ASIs and AWCs that underlies the foundational observations in figures 1 and 2 is not sufficiently characterized.
- (4) The assertion that 'starvation reshapes AWC responses from deterministic to stochastic' is not clearly supported by the data. AWC neurons seem capable of showing different responses to thermal stimuli, and the probabilities associated with these responses change after fasting. The different kinds of responses are seen under basal and fasted conditions.

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